

大数据场景下如何使用Volcano

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01

Spark场景使用Volcano



资源死锁

例如:

- Spark集群有2个CPU资源
- 每个Spark作业请求2个CPU资源，耗时10秒结束。

问题:

同时提交3个Spakr作业，需要多久运行完成?

K9s v0.25.18 minikube:minikube:v1.23.3 0%::0%								
Pods(default) [4]								
NAME↑	PF	READY	RESTARTS	STATUS	IP	NODE	A	
spark-pi-b3fbb080515190f2-exec-1	●	1/1	0	Running	172.17.0.7	minikube	1	
spark-test-app-7d00a63063404c9584eba1fd79d1a202-driver-1	●	0/1	0	Completed	172.17.0.6	minikube	3	
spark-test-app-7d00a63063404c9584eba1fd79d1a202-driver-2	●	0/1	0	Completed	172.17.0.6	minikube	3	
spark-test-app-7d00a63063404c9584eba1fd79d1a202-driver-3	●	1/1	0	Running	172.17.0.6	minikube	3	

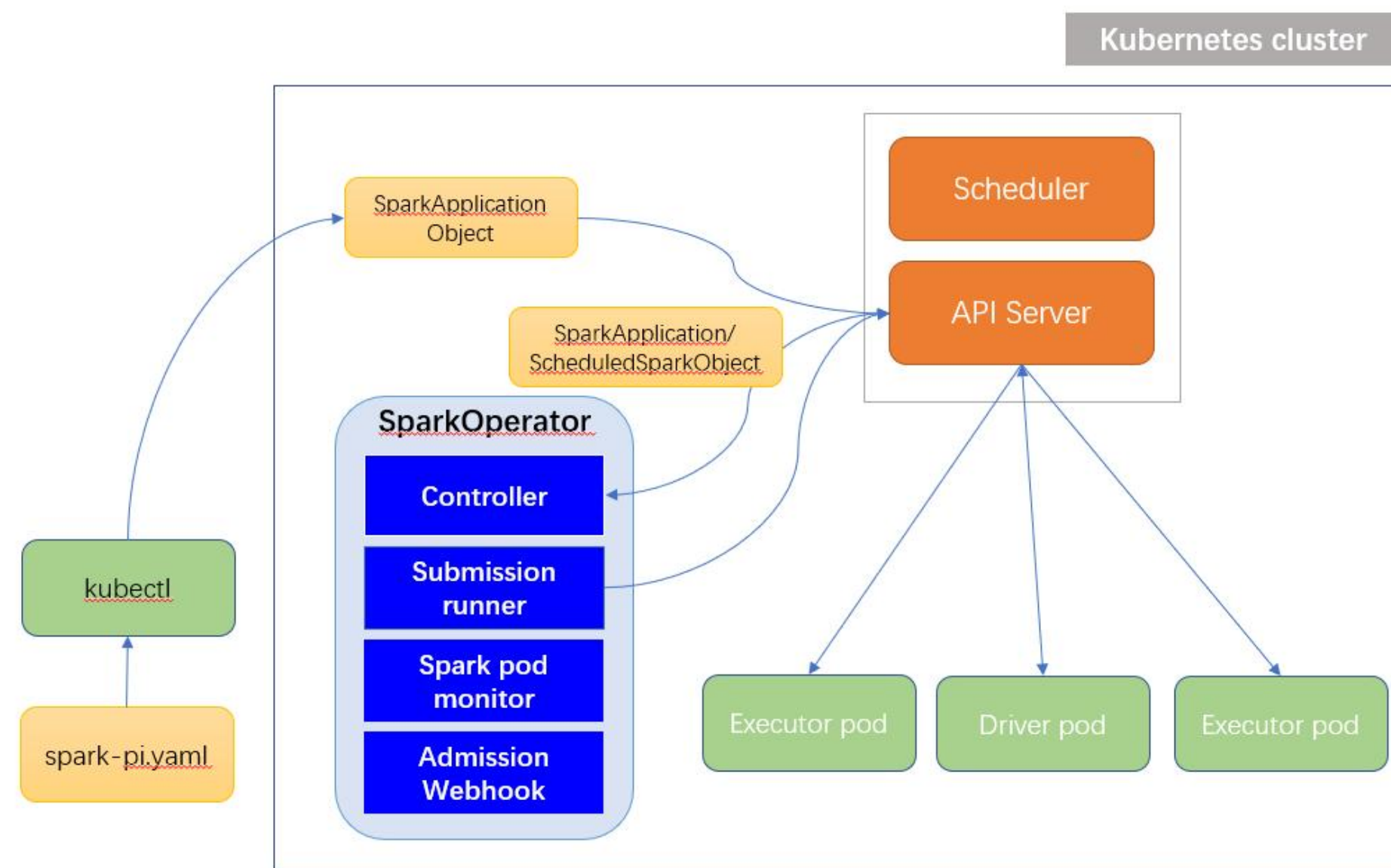
你认为30s

K9s v0.25.18 minikube:minikube:v1.23.3 0%::0%								
Pods(default) [5]								
NAME↑	PF	READY	RESTARTS	STATUS	IP	NODE	AGE	
spark-pi-00d842805153d35a-exec-1	●	0/0	0	Pending	n/a	n/a	5s	
spark-pi-9cdfb0805153d506-exec-1	●	0/0	0	Pending	n/a	n/a	5s	
spark-test-app-558a8ab1f8e94154bd4e93524ec5cbb2-driver-1	●	1/1	0	Running	172.17.0.7	minikube	12s	
spark-test-app-558a8ab1f8e94154bd4e93524ec5cbb2-driver-2	●	1/1	0	Running	172.17.0.6	minikube	12s	
spark-test-app-558a8ab1f8e94154bd4e93524ec5cbb2-driver-3	●	0/0	0	Pending	n/a	n/a	12s	

但是∞ !

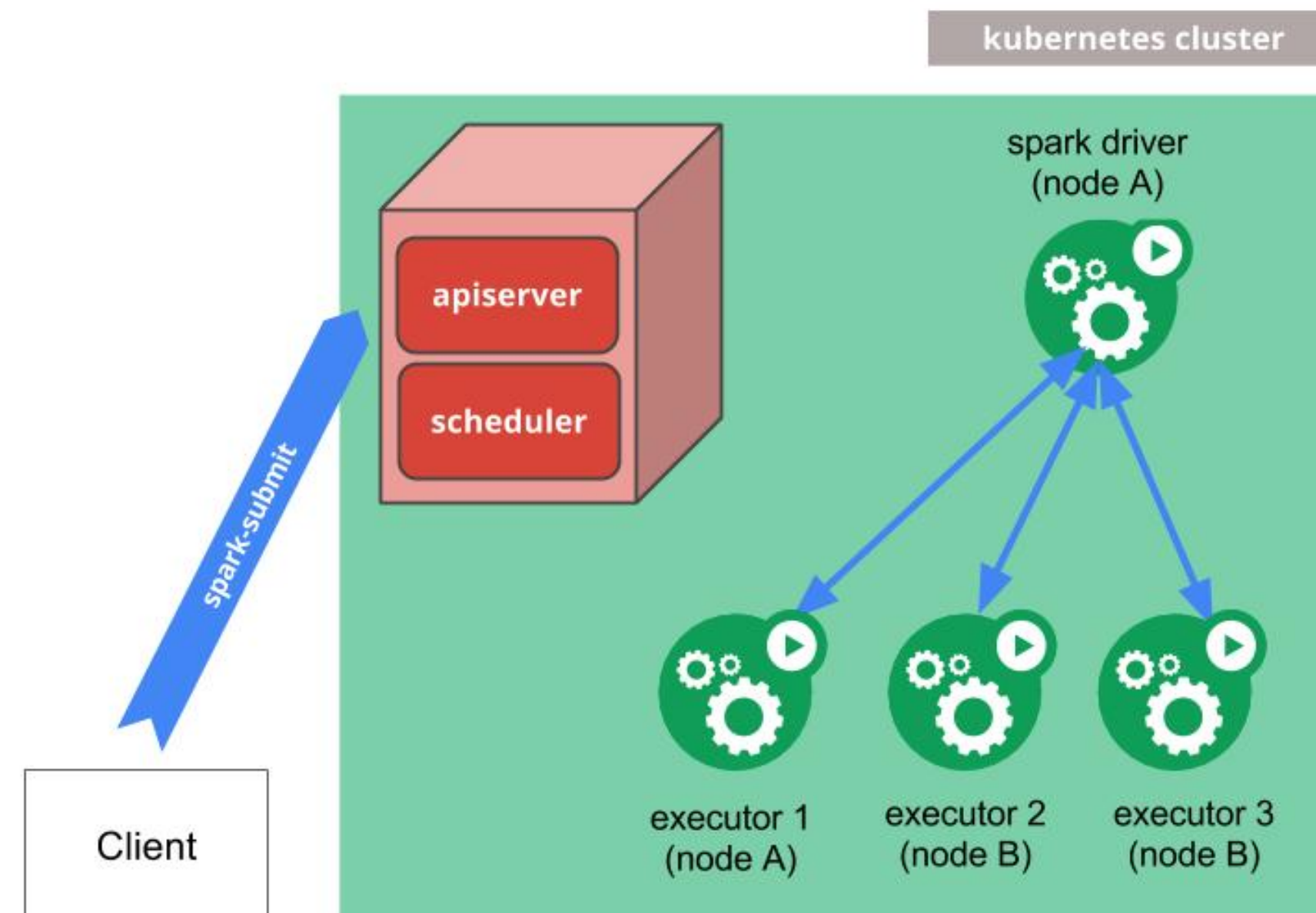


Spark在K8s上的运行方式



Spark on Kubernetes Operator

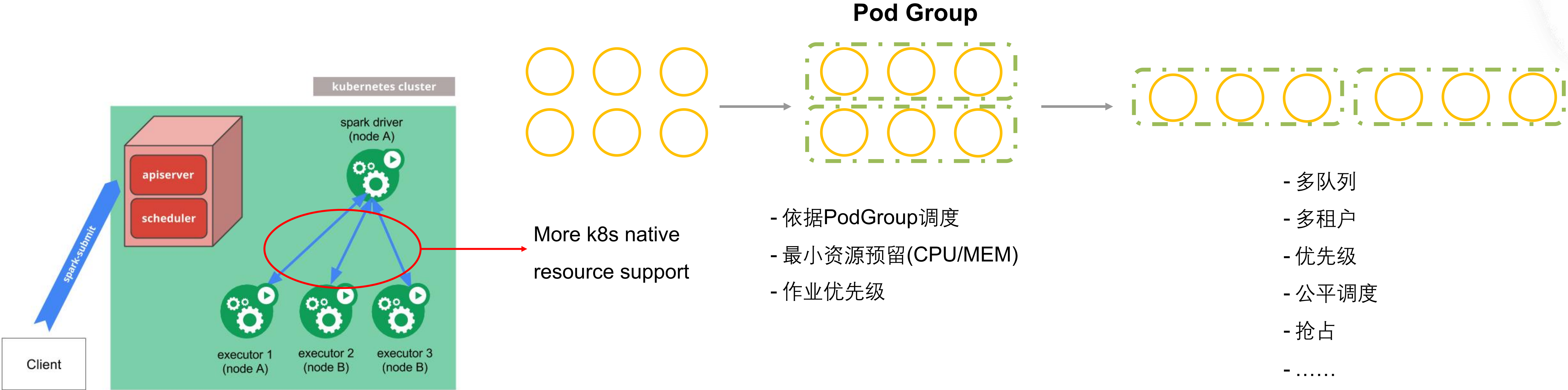
- Google opensource
- Submit: `kubectl apply -f *.yaml`
- Use yaml configure Kubernetes parameters



Native Spark App in K8s

- Apache Spark provides
- Submit: `./spark-submit`
- Use `--conf` configure Kubernetes parameters

Spark批量调度



1. Support replicaset/job API	RESOLVED	Holden Karau
2. Add the ability to specify a scheduler & queue	IN PROGRESS	Apache Spark
3. Support backing off dynamic allocation increases if resources are "stuck"	OPEN	Unassigned
4. Create a PodGroup with user specified minimum resources required	OPEN	Unassigned
5. Support for specifying executor/driver node selector	RESOLVED	Yikun Jiang
6. Support the Volcano Job API	OPEN	Unassigned

[SPARK-36057: Support volcano/alternative schedulers](#)

Native Spark Application

Spark Application

```
apiVersion: "sparkoperator.k8s.io/v1beta2"
kind: SparkApplication
metadata:
  name: spark-pi-1
spec:
  type: Scala
  mode: cluster
  image: "gcr.io/spark-operator/spark:v3.0.0"
  imagePullPolicy: IfNotPresent
  mainClass: org.apache.spark.examples.SparkPi
  mainApplicationFile: "local:///opt/spark/examples/jars/spark-examples_2.12-3.0.0.jar"
  sparkVersion: "3.0.0"
  batchScheduler: "volcano"
  ... ..
driver:
  cores: 1
  coreLimit: "1200m"
  memory: "512m"
  labels:
    version: 3.0.0
  annotations:
    scheduling.k8s.io/group-name: spark-pg-1
  ... ..
executor:
  cores: 1
  instances: 2
  memory: "512m"
  labels:
    version: 3.0.0
  annotations:
    scheduling.k8s.io/group-name: spark-pg-1
  ... ..
```

PodGroup

```
apiVersion: scheduling.volcano.sh/v1beta1
kind: PodGroup
metadata:
  name: spark-pg-1
spec:
  queue: default
  minMember: 1
  minResources:
    cpu: "3"
    memory: "1536m"
```



Spark Demo

Kubernetes version: 1.23
Spark operator version: v1beta2-1.3.4-3.1.1
Cluster free resource: 3C, 8G

Submit 3 Spark jobs, each job has 1 driver pod and 2 executor pods
Driver pod request 1C, 512m
Executor pod request 1C, 512m

```
root@w00564621-res-33064-1r3xs:~/peng/yaml/spark-demo#
```

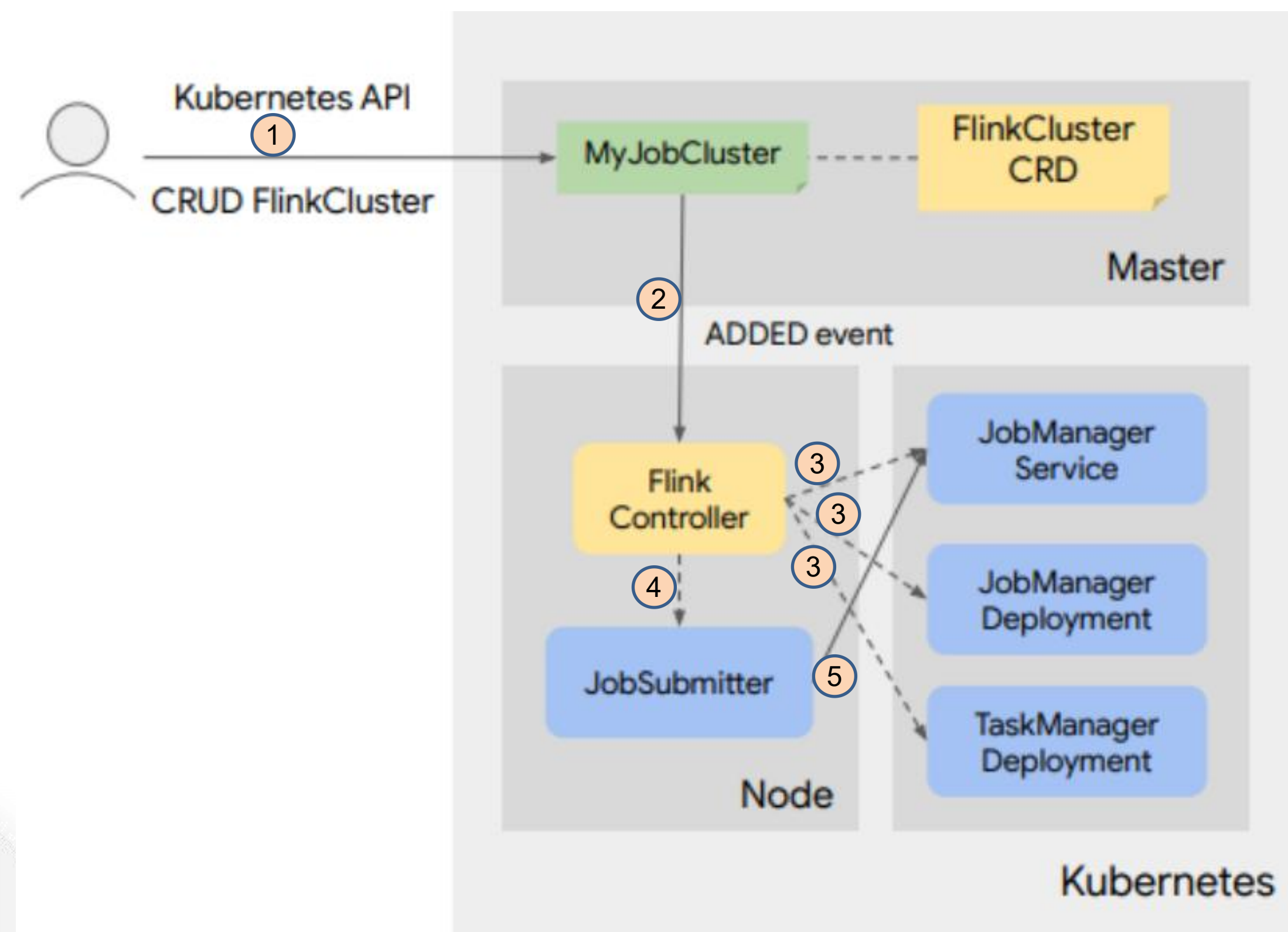
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02

Flink场景使用Volcano

Flink Operator 架构



1. 用户通过`kubectl apply -f myjobcluster.yaml` 命令提交 FlinkCluster spec 到API server。
2. Kubernetes将触发FlinkCluster ADDED 事件， Flink Controller将监控并处理该事件。
3. Flink Controller分析FlinkCluster CR后， 通过the API server 创建底层原始资源(JobManager service, JobManager deployment, TaskManager deployment)。
4. 当JobManager deployment, JobManager service, TaskManager deployment ready时候, the controller 创建 Flink job submitter。
5. Flink job submitter 提交作业到JobManager service 并且不断 查询作业状态, 当作业为结束或失败终结状态时， 终结job submitter。

Flink Demo

Kubernetes version: 1.15
Flink operator version: v1beta1-6
Cluster free resource: 3c, 8Gi

Submit 5 Flink jobs, each job has 1 jobmanager pod, 4 taskmanager pods and 1 job pod
Jobmanager pod request: 0.2C, 1Gi
Taskmanager pod request: 0.2C, 1Gi

```
[root@everest-storage-v15-9j8l2 flink-demo]#
```

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Github: <https://github.com/volcano-sh/volcano>



Slack Channel: <https://volcano-sh.slack.com/>



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